

Use Case Flow

Bank Management System

UC1 — Open new account (FR-1, FR-2)

Goal	Open a new customer bank account with valid KYC details and the required initial deposit.
Actor	Customer
Main flow	1. Customer selects to open a new account. 2. System prompts customer to enter KYC details (name, address, phone, ID proof) and initial deposit. 3. Customer enters the required details. 4. System validates the entered KYC details. 5. System validates that the initial deposit is at least ₹500. 6. System «includes» Generate unique 10-digit account number (UC2). 7. System creates the customer account. 8. System confirms the account has been created successfully.
Alternate flow	Initial deposit below ₹500 5a. System displays an error stating that the minimum initial deposit is ₹500. 5b. Customer re-enters the deposit amount and returns to step 5.
Exception flow	Invalid KYC details 4a. System displays an error and asks the customer to correct the details. 4b. Return to step 4.

UC2 — Generate unique 10-digit account number (FR-2)

Goal	Generate and assign a unique 10-digit account number for a new account.
Actor	System, triggered by UC1
Main flow	1. System generates a 10-digit account number. 2. System checks whether the generated account number already exists. 3. If the number is unique, the system assigns it to the new account. 4. Control returns to the triggering use case (UC1).
Alternate flow	Duplicate account number 2a. System rejects the duplicate number. 2b. System generates another 10-digit number and returns to step 2.
Exception flow	Account-number generation failure 3a. System reports the failure and account creation is not completed until a valid unique number is generated.

UC3 — View / edit / deactivate customer account (FR-3)

Goal	View, update, or deactivate an existing customer account.
Actor	Bank staff / Admin
Main flow	1. Staff/Admin selects a customer account. 2. System retrieves and displays the customer account details. 3. Staff/Admin chooses to view, edit, or deactivate the account.

	4. For edit, Staff/Admin updates the required account details. 5. System validates the updated details. 6. System saves the changes or confirms the account status change. 7. System confirms the requested operation.
Alternate flow	Invalid updated details 5a. System displays an error and asks Staff/Admin to correct the details. 5b. Return to step 5.
Exception flow	Customer account not found 2a. System displays an error and asks Staff/Admin to select another account or cancel.

UC4 — Withdraw money with minimum-balance validation (FR-4)

Goal	Complete a withdrawal only when the account retains the required ₹1,000 minimum balance.
Actor	Customer / Bank staff
Main flow	1. Customer/Staff selects to withdraw money. 2. System requests the withdrawal amount. 3. Customer/Staff enters the withdrawal amount. 4. System retrieves the available account balance. 5. System calculates the balance after the withdrawal. 6. System verifies that the resulting balance is at least ₹1,000. 7. System completes the withdrawal. 8. System «includes» Generate transaction ID and receipt (UC5). 9. System «includes» Log transaction details (UC7).
Alternate flow	Insufficient balance / minimum balance violation 6a. System rejects the withdrawal and displays an “Insufficient balance” message. 6b. Customer/Staff may enter a different amount.
Exception flow	Invalid withdrawal amount 3a. System displays an error and asks for a valid amount. 3b. Return to step 3.

UC5 — Generate transaction ID and receipt (FR-5)

Goal	Generate a unique transaction reference ID and receipt for every completed deposit or withdrawal.
Actor	System, triggered by deposit or withdrawal
Main flow	1. A deposit or withdrawal is completed successfully. 2. System generates a unique transaction reference ID. 3. System creates a receipt containing the transaction ID, date/time, transaction type, amount, and balance. 4. System displays or provides the receipt. 5. Control returns to the triggering transaction use case.
Alternate flow	Transaction ID generation failure 2a. System reports the failure and does not finalize the transaction until a unique reference ID can be generated.
Exception flow	Receipt generation failure 3a. System reports the receipt-generation failure and keeps the transaction record available in transaction history.

UC6 — Login and lock account after failed attempts (FR-6; NFR-3)

Goal	Authenticate a user and protect the account from repeated failed login attempts.
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Actor	Customer / Bank staff / Admin
Main flow	1. User selects to log in. 2. System requests username/account ID and password/PIN. 3. User enters credentials. 4. System verifies the credentials using the stored protected password/PIN value. 5. If credentials are valid, system allows access to the appropriate dashboard. 6. If credentials are invalid, system increments the failed-attempt count. 7. System checks whether 3 consecutive login attempts have failed. 8. If the count reaches 3, system locks the account for 15 minutes.
Alternate flow	Invalid credentials before the third failure 6a. System displays an authentication error. 6b. User may try again.
Exception flow	Third consecutive failed attempt 7a. System locks the account for 15 minutes. 7b. System denies further login until the lock period ends.

UC7 — Log deposit and withdrawal transactions / view history (FR-7)

Goal	Maintain an accurate transaction record for every completed deposit and withdrawal.
Actor	System, triggered by deposit or withdrawal; Customer / Staff for viewing
Main flow	1. A deposit or withdrawal is completed successfully. 2. System captures the timestamp, transaction type, amount, and post-transaction balance. 3. System stores the transaction in the transaction log. 4. Customer/Staff selects transaction history when required. 5. System retrieves and displays the recorded transactions.
Alternate flow	Save failure 3a. System displays an error and does not report the transaction as fully recorded until the save succeeds. 3b. System retries the save operation.
Exception flow	No transaction records found 5a. System displays an empty-history message.

UC8 — Generate admin report (FR-8; NFR-1, NFR-2)

Goal	Generate a report of total deposits, withdrawals, and active account count for a given date range.
Actor	Admin
Main flow	1. Admin selects to generate a report. 2. System requests the report type and date range. 3. Admin provides the report parameters. 4. System validates the report request and retrieves the relevant records. 5. System compiles the report containing total deposits, withdrawals, and active account count. 6. System generates the report within 10 seconds. 7. System displays or exports the report.
Alternate flow	Invalid date range / parameters 4a. System displays an error and asks Admin to correct the report parameters. 4b. Return to step 4.
Exception flow	No matching records

	4c. System displays an empty-report message.
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System-wide NFR constraints

NFR-1 Performance	95% of deposit and withdrawal transactions shall complete within 2 seconds, including under a load of up to 500 concurrent users.
NFR-2 Scalability	The system shall support at least 1,000 concurrent authenticated sessions without response time exceeding 4 seconds.
NFR-3 Security	Stored passwords and PINs shall use a one-way hashing algorithm such as bcrypt.
NFR-4 Data Integrity	The system shall reject transactions that would result in a negative balance or a duplicate account number.